THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 1

Chapter 12: **Surface Areas and Volumes** | Class 10 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(d)** 220 cm²	$2 \times \frac{22}{7} \times 7 \times 5$; CSA $= 220$ cm^2
2	**(c)** 1540 cm³	$\frac{22}{7} \times 7^2 \times 10$; Volume $= 1540$ cm^3
3	**(a)** 25 cm	$l^2 = 7^2 + 24^2 = 625$; $l = 25$ cm
4	**(c)** 550 cm²	CSA $= \pi rl$; CSA $= \frac{22}{7} \times 7 \times 25 = 550$ cm^2
5	**(c)** 616 cm²	$4 \times \frac{22}{7} \times 7^2$; Surface area $= 616$ cm^2
6	**(b)** 38808 cm³	$\frac{4}{3} \times \frac{22}{7} \times 21^3$; Volume $= 38808$ cm^3
7	**(a)** 462 cm²	$3 \times \frac{22}{7} \times 7^2$; TSA $= 462$ cm^2
8	**(b)** 880 cm²	$2 \times \frac{22}{7} \times 14 \times 10$; CSA $= 880$ cm^2
9	**(d)** 3080 cm³	$\frac{22}{7} \times 14^2 \times 5$; Volume $= 3080$ cm^3
10	**(b)** 50 cm	$l^2 = 14^2 + 48^2 = 2500$; $l = 50$ cm

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Total surface area $= 2\pi r(r+h)$; $2 \times \frac{22}{7} \times 7 \times (7+10)$ (1); TSA $= 748$ cm^2 (1).	1+1
12	Volume $= \frac{2}{3}\pi r^3$; $\frac{2}{3} \times \frac{22}{7} \times 10.5^3$ (1); Volume $= 2425.5$ cm^3 (1).	1+1
13	TSA $= 2(lb + bh + hl)$; $2(8 \times 5 + 5 \times 3 + 3 \times 8)$ (1); TSA $= 158$ cm^2 (1).	1+1
14	Total surface area $= 2\pi r(r+h)$; $2 \times \frac{22}{7} \times 14 \times (14+6)$ (1); TSA $= 1760$ cm^2 (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Volume $= \frac{1}{3}\pi r^2 h$ (1); $\frac{1}{3} \times \frac{22}{7} \times 7^2 \times 24$ (1); Volume $= 1232$ cm^3 (1).	1+1+1
16	Slant height $l = 25$ cm (1); TSA $= \pi r(l+r)$ (1); TSA $= \frac{22}{7} \times 7 \times (25+7) = 704$ cm^2 (1).	1+1+1
17	TSA $= 6a^2$ (1); $a^2 = 9$, so $a = 3$ cm (1); Volume $= a^3 = 27$ cm^3 (1).	1+1+1
	OR: Volume $= \frac{1}{3}\pi r^2 h$ (1); $\frac{1}{3} \times \frac{22}{7} \times 14^2 \times 48$ (1); Volume $= 9856$ cm^3 (1).	1+1+1
18	Slant height $l = 50$ cm (1); TSA $= \pi r(l+r)$ (1); TSA $= \frac{22}{7} \times 14 \times (50+14) = 2816$ cm^2 (1).	1+1+1
	OR: TSA $= 6a^2$ (1); $a^2 = 16$, so $a = 4$ cm (1); Volume $= a^3 = 64$ cm^3 (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Volume of cuboid $= n\times$ volume of a cube (2); $n = \frac{24\times12\times6}{3^3}$ (2); $n = 64$ (1).	2+2+1
	OR: Volume of big sphere $= n\times$ volume of a small sphere (2); $n = \frac{\frac{4}{3}\pi(6)^3}{\frac{4}{3}\pi(2)^3} = \left(\frac{6}{2}\right)^3$ (2); $n = 27$ (1).	2+2+1
20	Volume of cuboid $= n\times$ volume of a cube (2); $n = \frac{24\times12\times12}{6^3}$ (2); $n = 16$ (1).	2+2+1
	OR: Volume of big sphere $= n\times$ volume of a small sphere (2); $n = \frac{\frac{4}{3}\pi(9)^3}{\frac{4}{3}\pi(3)^3} = \left(\frac{9}{3}\right)^3$ (2); $n = 27$ (1).	2+2+1

Mistakes to avoid:

- Counting the joined faces in the total surface area of a combination of solids (e.g. adding the base of a cone resting on a hemisphere).
- Using the vertical height instead of the *slant* height in the CSA and TSA of a cone.
- Forgetting the $\frac{1}{3}$ in the volume of a cone or using $2\pi r^2$ for the surface area of a full sphere.
- Assuming surface area stays constant when a solid is melted and recast; only the volume is conserved.
- Mixing units (cm, m, mm) or writing cm instead of cm^2/cm^3 in the answer.
- Using diameter in place of radius while substituting into a formula.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 2

Chapter 12: **Surface Areas and Volumes** | Class 10 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(d)** 2200 cm²	CSA $= \pi rl$; CSA $= \frac{22}{7} \times 14 \times 50 = 2200$ cm^2
2	**(b)** 2464 cm²	$4 \times \frac{22}{7} \times 14^2$; Surface area $= 2464$ cm^2
3	**(c)** 310464 cm³	$\frac{4}{3} \times \frac{22}{7} \times 42^3$; Volume $= 310464$ cm^3
4	**(a)** 1848 cm²	$3 \times \frac{22}{7} \times 14^2$; TSA $= 1848$ cm^2
5	**(b)** 1056 cm²	$2 \times \frac{22}{7} \times 21 \times 8$; CSA $= 1056$ cm^2
6	**(b)** 2772 cm³	$\frac{22}{7} \times 21^2 \times 2$; Volume $= 2772$ cm^3
7	**(d)** 35 cm	$l^2 = 21^2 + 28^2 = 1225$; $l = 35$ cm
8	**(c)** 2310 cm²	CSA $= \pi rl$; CSA $= \frac{22}{7} \times 21 \times 35 = 2310$ cm^2
9	**(d)** 5544 cm²	$4 \times \frac{22}{7} \times 21^2$; Surface area $= 5544$ cm^2
10	**(b)** 4851 cm³	$\frac{4}{3} \times \frac{22}{7} \times 10.5^3$; Volume $= 4851$ cm^3

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Volume $= \frac{2}{3}\pi r^3$; $\frac{2}{3} \times \frac{22}{7} \times 21^3$ (1); Volume $= 19404$ cm^3 (1).	1+1
12	TSA $= 2(lb + bh + hl)$; $2(12 \times 7 + 7 \times 4 + 4 \times 12)$ (1); TSA $= 320$ cm^2 (1).	1+1
13	Total surface area $= 2\pi r(r + h)$; $2 \times \frac{22}{7} \times 21 \times (21 + 4)$ (1); TSA $= 3300$ cm^2 (1).	1+1
14	Volume $= \frac{2}{3}\pi r^3$; $\frac{2}{3} \times \frac{22}{7} \times 31.5^3$ (1); Volume $= 65488.5$ cm^3 (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Volume $= \frac{1}{3}\pi r^2 h$ (1); $\frac{1}{3} \times \frac{22}{7} \times 21^2 \times 28$ (1); Volume $= 12936$ cm^3 (1).	1+1+1
16	Slant height $l = 35$ cm (1); TSA $= \pi r(l + r)$ (1); TSA $= \frac{22}{7} \times 21 \times (35 + 21) = 3696$ cm^2 (1).	1+1+1
17	TSA $= 6a^2$ (1); $a^2 = 25$, so $a = 5$ cm (1); Volume $= a^3 = 125$ cm^3 (1).	1+1+1
	OR: Volume $= \frac{1}{3}\pi r^2 h$ (1); $\frac{1}{3} \times \frac{22}{7} \times 21^2 \times 72$ (1); Volume $= 33264$ cm^3 (1).	1+1+1
18	Slant height $l = 75$ cm (1); TSA $= \pi r(l + r)$ (1); TSA $= \frac{22}{7} \times 21 \times (75 + 21) = 6336$ cm^2 (1).	1+1+1
	OR: TSA $= 6a^2$ (1); $a^2 = 36$, so $a = 6$ cm (1); Volume $= a^3 = 216$ cm^3 (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Volume of cuboid $= n\times$ volume of a cube (2); $n = \frac{24 \times 18 \times 9}{3^3}$ (2); $n = 144$ (1).	2+2+1
	OR: Volume of big sphere $= n\times$ volume of a small sphere (2); $n = \frac{\frac{4}{3}\pi(12)^3}{\frac{4}{3}\pi(4)^3} = \left(\frac{12}{4}\right)^3$ (2); $n = 27$ (1).	2+2+1
20	Volume of cuboid $= n\times$ volume of a cube (2); $n = \frac{24 \times 18 \times 18}{6^3}$ (2); $n = 36$ (1).	2+2+1
	OR: Volume of big sphere $= n\times$ volume of a small sphere (2); $n = \frac{\frac{4}{3}\pi(10)^3}{\frac{4}{3}\pi(2)^3} = \left(\frac{10}{2}\right)^3$ (2); $n = 125$ (1).	2+2+1

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- Forgetting the $\frac{1}{3}$ in the volume of a cone or using $2\pi r^2$ for the surface area of a full sphere.
- Assuming surface area stays constant when a solid is melted and recast; only the volume is conserved.
- Mixing units (cm, m, mm) or writing cm instead of cm^2/cm^3 in the answer.
- Using diameter in place of radius while substituting into a formula.

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ANSWER KEY & MARKING SCHEME SET 3

Chapter 12: **Surface Areas and Volumes** | Class 10 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(d)** 4158 cm²	$3 \times \frac{22}{7} \times 21^2$; TSA $= 4158$ cm^2
2	**(c)** 528 cm²	$2 \times \frac{22}{7} \times 7 \times 12$; CSA $= 528$ cm^2
3	**(d)** 462 cm³	$\frac{22}{7} \times 7^2 \times 3$; Volume $= 462$ cm^3
4	**(b)** 29 cm	$l^2 = 21^2 + 20^2 = 841$; $l = 29$ cm
5	**(a)** 1914 cm²	CSA $= \pi r l$; CSA $= \frac{22}{7} \times 21 \times 29 = 1914$ cm^2
6	**(d)** 9856 cm²	$4 \times \frac{22}{7} \times 28^2$; Surface area $= 9856$ cm^2
7	**(b)** 130977 cm³	$\frac{4}{3} \times \frac{22}{7} \times 31.5^3$; Volume $= 130977$ cm^3
8	**(b)** 7392 cm²	$3 \times \frac{22}{7} \times 28^2$; TSA $= 7392$ cm^2
9	**(b)** 880 cm²	$2 \times \frac{22}{7} \times 35 \times 4$; CSA $= 880$ cm^2
10	**(b)** 1848 cm³	$\frac{22}{7} \times 14^2 \times 3$; Volume $= 1848$ cm^3

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	TSA $= 2(lb + bh + hl)$; $2(15 \times 10 + 10 \times 6 + 6 \times 15)$ (1); TSA $= 600$ cm^2 (1).	1+1
12	Total surface area $= 2\pi r(r + h)$; $2 \times \frac{22}{7} \times 35 \times (35 + 2)$ (1); TSA $= 8140$ cm^2 (1).	1+1
13	Volume $= \frac{2}{3}\pi r^3$; $\frac{2}{3} \times \frac{22}{7} \times 42^3$ (1); Volume $= 155232$ cm^3 (1).	1+1
14	TSA $= 2(lb + bh + hl)$; $2(20 \times 9 + 9 \times 5 + 5 \times 20)$ (1); TSA $= 650$ cm^2 (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Volume $= \frac{1}{3}\pi r^2 h$ (1); $\frac{1}{3} \times \frac{22}{7} \times 21^2 \times 20$ (1); Volume $= 9240$ cm^3 (1).	1+1+1
16	Slant height $l = 29$ cm (1); TSA $= \pi r(l + r)$ (1); TSA $= \frac{22}{7} \times 21 \times (29 + 21) = 3300$ cm^2 (1).	1+1+1
17	TSA $= 6a^2$ (1); $a^2 = 49$, so $a = 7$ cm (1); Volume $= a^3 = 343$ cm^3 (1).	1+1+1
	OR: Volume $= \frac{1}{3}\pi r^2 h$ (1); $\frac{1}{3} \times \frac{22}{7} \times 35^2 \times 12$ (1); Volume $= 15400$ cm^3 (1).	1+1+1
18	Slant height $l = 37$ cm (1); TSA $= \pi r(l + r)$ (1); TSA $= \frac{22}{7} \times 35 \times (37 + 35) = 7920$ cm^2 (1).	1+1+1
	OR: TSA $= 6a^2$ (1); $a^2 = 64$, so $a = 8$ cm (1); Volume $= a^3 = 512$ cm^3 (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Volume of cuboid $= n\times$ volume of a cube (2); $n = \frac{24 \times 24 \times 6}{3^3}$ (2); $n = 128$ (1).	2+2+1
	OR: Volume of big sphere $= n\times$ volume of a small sphere (2); $n = \frac{\frac{4}{3}\pi(8)^3}{\frac{4}{3}\pi(2)^3} = \left(\frac{8}{2}\right)^3$ (2); $n = 64$ (1).	2+2+1
20	Volume of cuboid $= n\times$ volume of a cube (2); $n = \frac{24 \times 24 \times 18}{6^3}$ (2); $n = 48$ (1).	2+2+1
	OR: Volume of big sphere $= n\times$ volume of a small sphere (2); $n = \frac{\frac{4}{3}\pi(15)^3}{\frac{4}{3}\pi(5)^3} = \left(\frac{15}{5}\right)^3$ (2); $n = 27$ (1).	2+2+1

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